GOURAV DAS

AI/ML & GenAI Engineer (RAG Systems | LLMs | Applied ML)

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PROFILE SUMMARY

AI/ML & GenAI Engineer with hands-on experience designing and deploying production-grade RAG systems, explainable ML models, and high-throughput APIs. Architected **DocuMind AI** (hybrid RAG with BM25 + ChromaDB Gemini 3072-dim embeddings, Reciprocal Rank Fusion, and faithfulness guardrails) and **Swachh.ai** (serverless regulatory decision engine on AWS ECS Fargate). Rigorous approach to model evaluation: eliminating data leakage, optimizing clinical recall (84.9%), and ensuring algorithmic explainability (SHAP). AWS Certified Cloud Practitioner. Seeking an entry-level GenAI / LLM or AI/ML Engineer role.

TECHNICAL SKILLS

• **GenAI & RAG:** LangChain, Retrieval-Augmented Generation (RAG), Hybrid Retrieval (BM25 + Dense Vectors, Reciprocal Rank Fusion), ChromaDB, Amazon Bedrock, Google Gemini API & 3072-dim Embeddings, Faithfulness & Hallucination Guardrails, Prompt Engineering

• **Machine Learning & XAI:** Scikit-Learn (Pipelines, ColumnTransformer, Cross-Validation), SHAP (TreeExplainer), Logistic Regression, Random Forest, SVM, KNN, TF-IDF, Class Imbalance Mitigation, Feature Engineering, Recall / Sensitivity, ROC-AUC

• **Backend & Cloud:** FastAPI, Next.js 16, React 19, Flask, Pydantic v2, Uvicorn, Docker, Nginx, SQLite, RESTful APIs, AWS (ECS Fargate, EC2, S3, CloudWatch, VPC, IAM) — AWS Certified Cloud Practitioner

• **Languages & Core CS:** Python, JavaScript (ES6+), TypeScript, SQL (PostgreSQL, SQLite); Data Structures & Algorithms, OOP, DBMS (actively practicing via NeetCode 150); Exposure to Big Data (Hadoop, HDFS, Spark)

FEATURED PROJECTS

DocuMind AI — Production-Grade RAG Document Platform — FastAPI, React 19, LangChain, ChromaDB, Gemini | Live Demo → Oct 2026

- Architected a modular RAG platform for document-grounded Q&A over complex PDFs with verifiable page citations using an asynchronous FastAPI backend (Pydantic v2) and React 19 interface.
- Engineered **Hybrid Retrieval** combining ChromaDB dense vector search (Gemini 3072-dim embeddings) and BM25 sparse keyword search via **Reciprocal Rank Fusion (RRF)**, surfacing exact keywords (acronyms, IDs) and semantic context; implemented an automated **Faithfulness Guardrail** scoring response entailment and a SQLite cache serving repeat queries in **<50 ms**; deployed on AWS EC2 behind Nginx.

Swachh.ai — Cloud-Native Regulatory Decision RAG — AWS ECS Fargate, Next.js 16, FastAPI, ChromaDB | Live App → Oct 2026

- Built solo for **Environmental Hacks (Bharat Builds Tour — WeMakeDevs & AWS)**: a location-aware decision engine providing legally grounded waste disposal pathways from Indian environmental statutory frameworks (SWM 2016, Battery Rules 2022).
- Designed a **3-tier reliability engine**: primary Gemini structured JSON extraction (strict Pydantic schema), exponential backoff retry, and deterministic ChromaDB fallback (all-MiniLM-L6-v2 ONNX embeddings) guaranteeing zero ungrounded guidance; deployed serverless on **AWS ECS Fargate, S3, ECR, and CloudWatch** with zero idle VM compute costs.

Fake News Detection AI — IIT Patna Candidate-Shortlisting — Scikit-Learn, TF-IDF, Streamlit | Live Demo → Jul 2026

- Developed an end-to-end NLP classification pipeline with a stratified 70:10:20 train/validation/test split and fixed random seeds for strict experimental reproducibility.
- Attained **98.57% accuracy, 98.63% weighted F1-score, and 0.999 AUC-ROC** on an unseen test set of 8,980 articles; deployed as an interactive Streamlit web application with real-time prediction confidence breakdown.

Explainable Heart Disease Decision Support System — B.Tech Capstone — Scikit-Learn, SHAP, Flask | Live Demo → 2024 – 2025

- Prevented critical train/test contamination by deduplicating the Cleveland dataset (**723 repeated entries removed down to 302 unique patients**); evaluated 5 classification models in Scikit-Learn pipelines with StandardScaler.
- Prioritized clinical recall (sensitivity) to minimize deadly false negatives: Logistic Regression achieved **84.9% test recall and 87.1% ROC-AUC** (SVM led 5-fold CV accuracy at 85.0%); integrated **Explainable AI (SHAP TreeExplainer)** for patient-specific feature attribution in a Flask web application with PDF reports.

EXPERIENCE & TRAINING

ML Intern — LearnDepth Academy LLP — Credit Risk & Model Complexity API (ML-T2-090) | Live Demo → Aug 2026 – Sep 2026

- Engineered an end-to-end Credit Risk REST API (Flask, Scikit-Learn Pipeline with ColumnTransformer, StandardScaler, OneHotEncoder, joblib) on OpenML German Credit benchmark.
- Benchmarked model complexity vs. business outcomes: demonstrated simple Logistic Regression matched Random Forest at 79.5% accuracy but caught **+10% more defaulting applicants (54% vs 44% recall)**, ran **14.6x faster in inference**, and provided mathematical linear-coefficient transparency required for ECOA/FCRA regulatory compliance.

AWS re/Start Cloud Trainee — Tata Strive — Core AWS & Linux Infrastructure Jun 2026 – Oct 2026

- Provisioned and configured core AWS cloud services (VPC, EC2, S3, IAM, Security Groups), Linux administration, and network routing in an enterprise-aligned, project-driven curriculum.

Big Data Analytics Bootcamp — NIELIT Kolkata (Govt. of India) — Big Data Technologies Feb 2026

- Completed a 90-hour government-sponsored program covering distributed Big Data systems (Hadoop, HDFS, Spark, Hive) and applied storage operations on Linux.

EDUCATION

B.Tech, Computer Science & Engineering — Future Institute of Engineering and Management (MAKAUT) | **CGPA: 7.56** 2026

Higher Secondary (Science, Class XII), Kendriya Vidyalaya Fort William, Kolkata (CBSE) — 2021 | 76.00% • Secondary (Class X) — 2018 | 61.20%

CERTIFICATIONS & HACKATHONS

• **AWS Certified Cloud Practitioner** — Amazon Web Services (Oct 2026) • **Building GenAI with Amazon Bedrock** — AWS Training (Aug 2026)
• **Environmental Hacks (Bharat Builds Tour)** — WeMakeDevs & AWS (Oct 2026) • **Machine Learning Using Python** — Ardent Computech (2023)